Figure 1

Cyberbullying Studio3-Proposal

HDForge

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Cyberbullying

HDForge

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1 Overview

Today is the digital age, and the Internet has become an integral part of our lives, facilitating interaction and connection on an unprecedented scale. However, along with the benefits, it has also brought great challenges, notably the rise of. This project aims to address the pressing issue of cyberbullying, with a particular focus on "trolling" - a ubiquitous form of cyber violence. Because the Internet does not have the constraints of the traditional physical environment, individuals face new risks, including anonymity and the ease with which harmful content can be distributed. The prevalence of cyberbullying among adolescents underscores the urgency of the issue. By delving into the conditions and mechanisms of cyberbullying, we seek to understand its complexities and develop strategies to mitigate its harmful effects. Given the harmful effects of cyberbullying on individuals and society, addressing this issue is critical to creating a safer and more inclusive online environment.

2 Background

The Internet has become deeply integrated into modern life, and people are more likely to interact with strangers than ever before, which has triggered the emergence of new modes of interaction. While the "conflictual tensions" associated with violence in the physical environment make violence less likely to occur in the traditional sense (Collins, 2009), the cyber-environment essentially removes the "conflictual tensions", making the Internet a breeding ground for violence.

Bullying consists of three key elements: intentional hurtful action, repeated over time, and involving an imbalance of power ((Eleni, 2014). Traditionally confined to physical spaces like schools and workplaces, the rise of the internet has given birth to a new form of bullying - cyberbullying. Operating through digital mediums such as the internet and electronic devices, cyberbullying inflicts harm through text or images, affecting a shocking proportion of internet users, ranging from 10% to 40% (Whittaker & Kowalski, 2015). The consequences can range from temporary distress to tragically, even suicide. Teenagers are particularly affected, with 10-42% of teens in the UK, Australia, Canada and US reporting about cyberbullying (Kraft, 2006). The reason why teenagers are more susceptible is reflected in two aspects. Firstly, they have low judgement ability and limited ability to self-regulate their emotions and behaviours (Hamm, 2015), so they are more likely to be the target of cyberbullying and are more susceptible to the influence of bad words and values. Secondly, teenagers are heavy Internet users, with a high percentage of 95% of teenagers using the Internet, and 81% of them using social media (Pew Research Centre, 2018), so it can be said that their lives are already inseparable

from the Internet. Consequently, targeting youth for intervention programs becomes imperative.

Wang and Yun (2023) analyse the conditions and mechanisms of cyber violence from three perspectives: social cognitive mechanism, social emotional mechanism and social value mechanism. In addition to the fact that negative emotions are more likely to infect each other, and antagonistic situations are more likely to be generated in the internet, the fragmentation of information reinforces labelling, and people are more likely to focus on the bad traits in a person rather than judging a person as a whole. In addition, the anonymity (Papacharissi, 2016) and immediacy of cyberspace makes it more difficult for victims to identify their aggressors and collect evidence, resulting in low costs of violating the law and high costs of defending their rights. The egalitarian nature of the internet further facilitates bullying by providing equal voice to all.

The affordances of the web, including accessibility affordance (allow perpetrator to reach their targets beyond many constraints such as time and space), information retrieval affordance (a lot of personal information is displayed on the web and is not subject to any access restrictions), editability affordance (perpetrator can edit information related to bullying), and association affordance (the extent to which users interact with other users on social networking platforms), create conducive environments for various forms of cyberbullying (Tommy et al., 2019).

Because of the complexity of the Internet, cyberviolence takes many forms, including flaming, harassment, denigration, impersonation, outing and trickery, exclusion, cyberstalking, happy-slapping (Kowalski et al., 2008) and trolling (Developed on the web in the 1990's). When confronted with various forms of online violence, our group has decided to focus on trolling. Trolling is slang for the act of intentionally sending provocative or offensive words in order to arouse the anger of others on the Internet, and its corresponding noun is troll, which refers to the person who commits the act. Trolling is usually intentional, but there are unintentional cases where the user may just want to refute some points, they are stubborn and unwilling to recognize and accept others' points of view, which can be caused by confirmation bias. Confirmation bias is the behaviour that occurs when people are only willing to accept information that supports their point of view and ignore information that conflicts with it. This bias more easily occurs online, especially as the increasing sophistication of algorithmic technology puts people in an information cocoon, where they are only exposed to information they are comfortable with, which serves as an additional incentive for them not to accept opposing viewpoints. In most cases, however, trolling is a form of malicious behaviour, whereby trolls attempt to provoke bad feelings in others (PCMAG, 2009) and get people into pointless circular arguments (Turner, 2005), usually for entertainment purposes only (Buckels et al., 2014). Finding entertainment in conflict would have been understandable because it is a way of life for people, for example, people enjoy high-risk sports, action movies, violent computer games, etc., but these are simulated conflicts and trolling is real to attack people, which can cause irreversible damage to the individual to society (Hardaker, 2010).

3 Stakeholders

During the development stage, anyone who might be affected by our actions is a stakeholder we need to engage with, especially teenagers, who are our target users. Our background research indicates that they are the most likely to get emotionally affected by ‘trolling’ because teenagers have lower self-control, which makes them the best target or the abuser of cyberbullying incidents. Other stakeholders include potential counsellors, such as schools and legal aid organisations, who have the most experience in dealing with teenagers and determining the systematic action we should take based on the situation. Another group of stakeholders that

might receive benefits from our project is the technology companies. The fact that our project aims to be used on social media platforms to prevent and educate users on trolling. This leads us to create an environment where everyone will feel safe and interact with other people online. That's why we have to work closely with those companies to prevent damage that might come out of our proposed solution or receive others' feedback to improve our solution.

4 Ethical Considerations

In accordance with Key Ethical Standards for UI and UX design (Designlab, 2024), the information regarding the relative merits of the initial potential solutions developed were explored. This information was then separated into negatives and positives, and visually organised in a Venn diagram *(refer to fig.1 below)*, taking into account things such as costs, ethical considerations, security and other risks and benefits to the stakeholders.

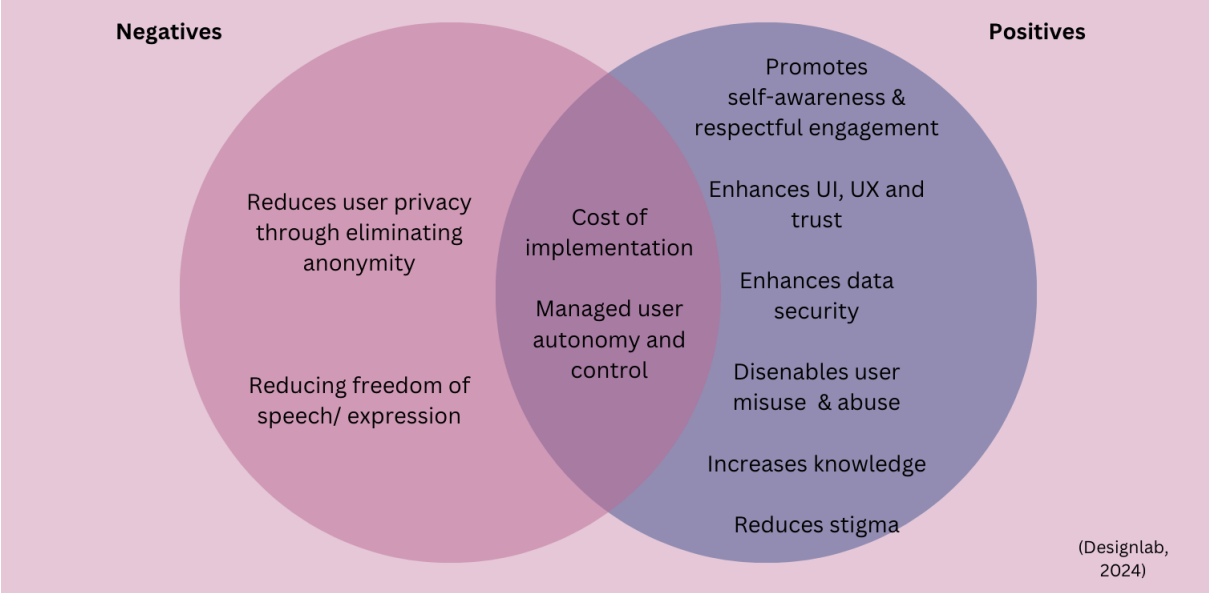

Figure 2
Shows a Venn diagram of the expected negative and positive impacts of the project's initial potential solutions for the problem scope.

The Venn diagram above highlights the positives and negatives of our initial potential solutions in response to the problem space of cyberbullying. While implementing these solutions may reduce user privacy and freedom of speech, they also promote awareness, accountability, and respectful engagement. Additionally, they can enhance user interaction, trust, and data security, while preventing misuse of online privileges and increasing knowledge about online safety. Overall, addressing cyberbullying requires a balanced and ethically sound approach that weighs the benefits and drawbacks of these solutions.

Following this, to ensure that the final solution is effective, fair, and respects the rights and well-being of all users, it was evident that several ethical considerations should be taken into account when designing a digital solution to address cyberbullying and in order to help determine the direction of our project. The key ethical considerations identified at this stage of the project are:

- Informed consent: While gaining users' consent is always a priority, particularly when the monitoring or analysing of users' personal data - such as their online activities and interactions with other users - is involved, it is critical that an informed

consent is obtained from users prior to the implementation of any digital solution. This is to ensure the purpose, scope, and potential risks associated with the solution have been clearly communicated, and users remain in control of opting in or out.

- Privacy and Data Protection: The digital solution must respect users' privacy rights and comply with data protection regulations. In terms of data collection, it must only collect necessary information for the solution to function effectively, making sure to securely store and process any personal data collected from users.
- Accessibility: It must be ensured that the digital solution is accessible to all users, including those with disabilities or limited access to technology. The final solution must be designed with inclusive features and provide sufficient support for users who may require assistance in using it effectively.

5 Initial Conceptual Model

The understanding of the problems surrounding cyberbullying, particularly trolling targeting teenagers, is critical for devising effective interventions. Cyberbullying, characterized by intentional and repeated hostile behaviour through digital platforms, poses significant psychological and emotional risks to its victims, often leading to distress, anxiety, and in extreme cases, self-harm or suicide. The pervasive nature of cyberbullying is compounded by the anonymity and immediacy of online interactions, which embolden perpetrators and make it challenging for victims to seek help or escape from the harassment.

Teenagers, being heavy users of digital communication platforms, are particularly susceptible to cyberbullying due to their still-developing coping mechanisms and vulnerability to peer pressure. The prevalence of cyberbullying is further exacerbated by the widespread accessibility of the internet, enabling bullies to target victims beyond physical boundaries and around the clock.

Understanding these challenges underscores the importance of developing comprehensive strategies to address cyberbullying effectively. It necessitates not only technological solutions such as AI-powered moderation tools and user-friendly reporting mechanisms but also educational initiatives that empower teenagers with the skills and knowledge to recognize and respond to cyberbullying. Psychological support services tailored specifically to teenagers affected by cyberbullying are crucial for providing emotional assistance and coping strategies. Moreover, community engagement efforts, legal frameworks, and research endeavours are essential components for creating a safer online environment for teenagers.

Based on the existing design ideas, I try to highlight their strengths and identify the areas for improvements through feedbacks.

- Technological Tools:
    - Strengths: The development of AI-powered moderation tools and user-friendly reporting mechanisms is a proactive step in combating cyberbullying efficiently.
    - Areas for Improvement: Ensure that AI algorithms are continuously refined to minimise false positives and negatives. Also, prioritise user testing and feedback to optimise reporting mechanisms for ease of use and accessibility.

- Educational Initiatives:

    - Strengths: Integrating cyberbullying awareness programs into school curriculums and creating interactive online resources empower teenagers with practical strategies for dealing with cyberbullying.
    - Areas for Improvement: Collaborate with educators and psychologists to ensure the relevance and effectiveness of educational materials. Regular evaluation and updates based on feedback and emerging research are essential.

- Psychological Support:
    - Strengths: Offering online counselling services and establishing safe online spaces provide valuable support to teenagers affected by cyberbullying.
    - Areas for Improvement: Ensure the accessibility, confidentiality, and quality of online counselling services. Provide clear guidelines for participation in online support groups to ensure safety and confidentiality.

- Community Engagement:
    - Strengths: Collaborating with influencers, celebrities, and community leaders can amplify awareness efforts and mobilise support for anti-cyberbullying campaigns.
    - Areas for Improvement: Ensure sustained engagement and impact beyond the duration of events or campaigns. Leverage social media and digital platforms for ongoing engagement and advocacy.

- Policy and Legal Frameworks:
    - Strengths: Advocating for stronger legal measures and policies demonstrates a commitment to addressing cyberbullying at a systemic level.
    - Areas for Improvement: Collaborate with legal experts and policymakers to draft comprehensive and enforceable legislation. Conduct public education campaigns to raise awareness about legal consequences of cyberbullying.

- Research and Evaluation:
    - Strengths: Conducting research and evaluation activities can provide valuable insights into the prevalence, impact, and effectiveness of interventions aimed at combating cyberbullying.
    - Areas for Improvement: Collaborate with academic institutions and research partners to design rigorous studies and prioritise research questions relevant to practitioners and policymakers.

6 Questions of Investigation

In the wake of the internet era, the advent of cyberbullying has emerged as a significant problem, presenting complex challenges to our team's project from legal, societal, and ethical perspectives. Cyberbullying encompasses a wide range of social phenomena, leading us to focus on trolling, a prevalent form of this disruptive behaviour. This focus necessitates the design and adoption of a suite of appropriate methods that not only leverage current research findings but also take into account the reality of today's internet interactions. Therefore, our approach requires an in-depth analysis of the motives behind online incivility, aiming to uncover gaps in our current understanding, especially concerning legal frameworks, psychological motivations, and social dynamics.

The diversity of international laws poses a significant challenge to our project. Moreno and Whitehill (2018) advocate for a strategy that adheres to the legal variations across nations while combating cyberbullying. This highlights the complexity and necessity of developing solutions that are flexible and applicable within various legal frameworks. However, aligning our solutions with the internet's inherent flexibility and anonymity introduces additional complexity. The Cybersecurity and Infrastructure Security Agency's (n.d.) work becomes a crucial reference, offering strategies for legal compliance to ensure our interventions are accepted publicly.

The research on the psychological motivations behind internet trolling reveals how online anonymity encourages such behaviours (Nitschinsk et al., 2022). This insight guides our exploration of how to balance reducing anonymity to deter trolling while maintaining digital media freedom, addressing the intricate moral and psychological dilemmas involved.

Cruz et al. (2018) explore the societal impacts of trolling, showing that it extends beyond individual experiences to threaten entire communities and exacerbate social divisions. Their research underscores the importance of interventions that make internet trolling sensitive to social dynamics, especially targeting marginalised groups. This calls for a deeper understanding of how to harness the dynamism of online platforms to positively impact social inclusivity, thus mitigating the divisive effects of trolling.

Furthermore, the advanced ethical considerations raised by Kowalski et al. (2021) become a significant aspect of our project, focusing on combating cyberbullying with user consent and data privacy. The challenge of ethically collecting data while effectively maintaining user safety and privacy protection prompts serious reflection.

To bridge the gaps in our knowledge and develop comprehensive solutions to mitigate the societal chaos caused by internet trolling, our project includes targeted research initiatives and strategies. We plan to conduct exploratory studies to investigate more deeply the relationship between trolling and anonymity, engage in social initiatives, understand the perspectives and needs of severely affected victims, and collaborate with legal and ethical experts. This ensures our solutions are grounded in both legal legitimacy and ethical constraints.

By building on the diversity of international laws and incorporating contemporary research insights, our project aims to contribute to a safer, more inclusive online environment. Addressing trolling requires not only technological innovation but also insight into legal, societal, and ethical dimensions. Only through continuous engagement and problem-solving across multiple dimensions can our solutions meet the dynamic needs of the internet era and effectively intervene in the phenomenon of internet trolling, gradually achieving the development of a comprehensive digital society.

7 User Research

To gather data to support the design direction for understanding the problem space of cyberbullying, particularly focusing on trolling and its impact on teenagers. several methods are employed in our project:
Literature Review: Extensive background reading and literature review have been conducted, covering academic research, industry reports, and expert opinions on cyberbullying, its prevalence, mechanisms, and impact. This literature review has provided a comprehensive understanding of the problem space and existing interventions.

Identification of Stakeholders: Key stakeholders, including teenagers (especially), educators, mental health professionals, legal aid organisations, and technology companies, have been identified. Understanding their perspectives and needs is crucial for designing effective solutions to address cyberbullying.
Ethical Considerations: Ethical considerations have been carefully examined, including privacy and data protection, informed consent, accessibility, and user autonomy. These considerations are vital for ensuring that the proposed solutions are fair, respectful of users' rights, and mitigate potential harms.
Initial Conceptual Model: An initial conceptual model has been developed, outlining the understanding of the problem space and potential solutions. This model highlights the importance of technological tools, educational initiatives, psychological support, community engagement, policy and legal frameworks, and research and evaluation in combating cyberbullying.

To further support the design direction and address the identified questions, the following steps could be taken:
User Research: Conduct interviews, surveys, or focus groups with stakeholders, including teenagers, educators, mental health professionals, and technology experts, to gain deeper insights into their experiences, needs, and preferences related to cyberbullying and potential solutions.
Prototyping and Mock-ups: Develop low-fidelity prototypes or mock-ups of proposed solutions, such as AI-powered moderation tools, educational resources, online counselling platforms, or community engagement campaigns. Gather feedback from stakeholders to iteratively refine and improve the design.
Observations and Field Studies: Observe online interactions and behaviours in relevant online communities or social media platforms frequented by teenagers. This could provide valuable insights into the dynamics of cyberbullying and inform the design of effective interventions.
Technology Spikes: Explore the feasibility of implementing specific technical solutions, such as AI algorithms for content moderation or user-friendly reporting mechanisms, through technology spikes or proof-of-concept studies.
By engaging in user research, prototyping, observations, and technology spikes, the design direction can be further informed and refined, leading to the development of more effective and user-centred solutions to combat cyberbullying.

8 Project Plan

To solve the problems identified in the conceptual model, we planned several activities for different stages of our project.
First of all, In Week 6, we will conduct user research and interviews, which will mainly investigate the needs of our stakeholders and our target users. In this way we can get into the minds of the users and gain a deeper understanding of what is required for our project to succeed. After that Week 7 involves determining essential functions through quantitative analysis since by Week 7 we would have gathered useful data from Week 6's research and interview. We can analyse those data and decide and select what possible functions we might want in our project that align with the final solution. By Week 8, we start prototyping and prepare for critique. During this time, we will investigate what framework will best suit our users, the investigation might also include consultation with our stakeholders such as schools to learn how to educate teenagers and technology companies to learn what are the platform policies. Weeks 9-12 focus on user experience and addressing ethical concerns. From the last investigation, we have a better idea in terms of education and platform policies, all of this can be used to

avoid any ethical concerns our project might bring to the people. In the meantime, we will prepare the final documentation and be ready to present our project in Week 13. Our team will have a regular Wednesday meeting at 5 pm on Zoom to ensure continuous improvement based on feedback. We also have our group Google Drive to record and keep track of the progress we made.

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